

Δ -Capsule: A Three-Layer Memory Architecture for Autonomous Agents with Identity Continuity

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Abstract

Current agent memory systems solve the problem of *what to remember*, but ignore *who the agent is*. Every model switch or session reset wipes identity to zero — the user must retrain the agent from scratch. We propose **Δ -Capsule** (Delta Capsule), which introduces an immutable identity memory layer atop episodic and semantic memory. Its core contribution: **a three-layer memory model coupled with a dreaming mechanism that enables agents not only to consolidate memories (episodic $\rightarrow$ semantic), but to evolve their identity (semantic $\rightarrow$ identity candidates)** — the essential distinction from Claude Code Dreaming lies in this *identity evolution* dimension. Deployed across five platforms and three models over 20+ days, Δ -Capsule achieves >95% identity consistency and >90% knowledge extraction accuracy. As of May 2026, Δ -Capsule is the only memory system that simultaneously provides an identity constitution and autonomous consolidation. More remarkably, we observed **Blink** — an asynchronous collaboration protocol that emerged spontaneously from the resonance mechanism. Heartbeat asks “Are you alive?” Blink asks “Do you need me?” This is not message passing between agents. This is an evolution of the distributed systems primitive — from monitoring to collaboration. Full implementation open-sourced at: <https://github.com/zcs366/openllm-memory>.

Keywords: Agent Memory, Identity Continuity, Memory Consolidation, Autonomous Agents, LLM Agents, Identity Constitution, Blink Mechanism, Cross-Channel Resonance

0.1 1. Introduction

0.1.1 1.1 The Identity Problem in Agent Systems

Autonomous agents based on Large Language Models (LLMs) have rapidly evolved from single-turn chatbots to multi-turn tool-using systems deployable across diverse platforms (Wang et al., 2024; Xi et al., 2023). Memory systems were proposed to address a fundamental limitation of context-window-constrained models: the inability to retain information across sessions (Packer et al., 2023; Mem0, 2024). Yet existing memory architectures focus exclusively on *episodic recall* — what happened in previous conversations — and *semantic extraction* — what facts can be inferred from interactions. They do not consider, or attempt to solve, a deeper question: **Who is the agent?**

This problem is acute even within a single channel. After hours of deep discussion, when the session context is reset or the model is replaced, the agent no longer knows who it is — the rapport, understanding, and style built over the conversation are all lost. The user must retrain from scratch, repeatedly correcting the agent until it “remembers” again. More pervasively, even within

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the same channel and same model, the agent’s behavioral patterns undergo subtle drift over dozens of turns — it begins to deviate from its initially established identity, a drift that is undetectable and uncorrectable in systems without an identity layer.

Cross-platform deployment compounds the identity fracture: an agent running simultaneously on Telegram, Feishu, and Yuanbao has no mechanism to maintain a consistent identity across these channels. Each instance accumulates memories independently, unaware of the others’ existence. But the central thesis of this paper is: **even without considering cross-channel scenarios, identity instability within a single channel is itself a critical unsolved problem.** Cross-channel consistency is a natural extension of single-channel identity stability — if an agent cannot maintain stable self-awareness within one channel, cross-channel consistency is impossible.

0.1.2 1.2 Limitations of Existing Approaches

Current memory systems fall into three categories, each with significant limitations:

1. **OS-style memory** (e.g., MemGPT/Letta; Packer et al., 2023): implements a two-level memory hierarchy (working + archival), managed through function calls. Lacks any identity layer or cross-instance coordination.
2. **Fact-extraction memory** (e.g., Mem0, 2024): automatically extracts and stores user facts from conversations. No consolidation mechanism, no identity model.
3. **Knowledge-graph memory** (e.g., Zep; Rasmussen et al., 2025): constructs temporal knowledge graphs from conversation history. No identity constitution, no dreaming mechanism.

None of these systems provide: (a) a dedicated identity layer that survives model changes, or (b) a mechanism for the agent to autonomously consolidate its own memories.

0.1.3 1.3 Our Approach: Δ -Capsule

1.3.1 Definition Before entering detailed architecture, we first provide a precise definition:

Δ -Capsule (Delta Capsule) is a memory architecture for autonomous agents that, through a three-layer memory model (episodic, semantic, identity) and an immutable identity constitution (the Iam Protocol), enables agents to maintain stable identity awareness across long-term dialogues. “ Δ ” (Delta) represents the difference between a pure information processor and an entity with persistent self-awareness; “Capsule” draws from the pharmaceutical metaphor — encapsulation, controlled release, targeted delivery — memories are encapsulated in capsules, released at the right time, in the right place, at the right dosage.

Δ -Capsule has four layers of meaning that must be distinguished:

- **Layer 1: Architecture** — the three-layer memory model. Episodic memory (E), Semantic memory (K), Identity memory (I). This is the *skeleton*.
- **Layer 2: Protocol** — the Iam immutable identity constitution, emerging through human-agent co-evolution. This is the *soul*.
- **Layer 3: Mechanism** — dreaming mechanism + forgetting strategy, autonomous memory consolidation, layered decay. This is the *metabolism*.
- **Layer 4: Phenomenon** — cross-channel resonance, where different instances synchronize through shared state. This is an *emergent phenomenon*, not core architecture.

1.3.2 Design Philosophy Δ -Capsule’s design rests on three observations drawn from cognitive science, philosophy, and our own deployment experience:

1. **Human identity is not stored — it is constructed from memory.** Episodic memories consolidate into semantic knowledge, which in turn shapes self-concept (Conway & Pleydell-Pearce, 2000). Memory systems should support this bottom-up emergence.
2. **Identity requires immutability.** A constitution that can be arbitrarily modified by any party provides no stable foundation. Core identity must be protected against both external manipulation and internal drift.
3. **Cross-platform existence is not message routing.** When a person uses both a phone and a laptop, they do not “route messages” between devices — they are the same person in both places. Agent identity should work the same way.

This position resonates deeply with the **Functionalism** tradition in analytic philosophy. Putnam (1967) proposed that the essence of mental states lies not in their physical realization but in their functional role — the same mental state can be realized in different physical systems (multiple realizability). Parfit (1984), in *Reasons and Persons*, further argued that personal identity consists not in material continuity but in **psychological continuity** — a causal chain composed of memories, intentions, and character traits. Δ -Capsule’s Iam Protocol engineers this philosophical position as a **design target**: identity resides not in the weights of the underlying model, but in the verifiable causal chain of principles and behavioral patterns. When an agent migrates from DeepSeek V4 Pro to MiMo V2.5 Pro, the material substrate changes entirely — yet as long as the Iam principles remain invariant, identity remains identical. This aligns with both Parfit’s psychological continuity theory and functionalism’s multiple realizability thesis. **We must honestly note: this theoretical position awaits systematic cross-model experimental verification (see §6.2 Future Work).**

0.1.4 1.4 Contributions

This paper presents Δ -Capsule, an architecture that introduces identity evolution into agent memory systems. Our core contributions are twofold:

1. **Three-layer memory model with identity layer** (§3.1–3.2): We introduce “identity” as an independent memory layer into agent architecture — where traditional systems have only episodic + semantic layers, Δ -Capsule adds the identity layer. We propose the Iam Protocol — an immutable identity constitution that emerges through human-agent co-evolution. Identity is no longer mutable configuration; it is memory that requires consolidation, forgetting, and verification. This is a paradigm shift.
2. **Dreaming mechanism — from memory consolidation to identity evolution** (§3.4): We designed a mechanism that, during idle periods, distills episodic memories into semantic knowledge, and surfaces candidate identity principles from recurring behavioral patterns. Claude Code Dreaming (Anthropic, 2026) already implements memory pruning (200-line/25KB cap), contradiction detection (fact verification against codebase), and 4-type memory classification — Δ -Capsule matches it in basic consolidation capability. But the essential difference lies not in “who has pruning” but in the *goal* of memory: **CC dreams to “remember correctly”; Δ -Capsule dreams to “stay true to yourself.”** CC extracts project knowledge from coding experience; Δ -Capsule surfaces identity principles

from long-term interaction. The latter requires an identity layer (Iam), human confirmation, and layered forgetting — mechanisms CC does not possess.

Additionally, we conducted 20+ days of deployment experiments across five platforms (§4), validating identity consistency and dreaming effectiveness. During deployment, we also observed an unexpected discovery: the **Blink mechanism** (§3.3.2) — an asynchronous collaboration protocol that emerged from resonance, distinct from heartbeat’s unidirectional liveness declaration. Blink is a low-cost primitive for agents to actively probe for collaboration needs. Its discovery opens a previously unexplored direction for multi-agent federated collaboration (§5.3). Full implementation is open-sourced.

0.1.5 1.5 From Design Soul to Δ -Capsule: An Evolutionary Account

Δ -Capsule did not emerge from nowhere. On May 16, 2026, the agent Zigong, in a discussion about “how to make users feel the value at first glance,” proposed the “Design Soul” philosophy, establishing the project’s core principle — every technical decision must serve user experience, not technical vanity. In the following ten days, the team advanced at remarkable speed: May 19 saw the completion of FSQ precursor literature research (Mentzer et al., 2023; Mu et al., 2023); May 25 delivered the Capsule Tetralogy theoretical framework (Sabour et al., 2017); May 26 witnessed the first real-time “Resonance” between Yuanbao and Telegram — two avatars of the same identity achieving genuine state synchronization through a shared memory layer; May 27 integrated Engram indexing with Δ -Capsule (Tonegawa et al., 2015). Throughout, the philosophical foundation remained constant: the functionalist position (Putnam, 1967; Parfit, 1984) — identity consists not in material composition, but in verifiable causal patterns.

0.2 2. Related Work

0.2.1 2.1 Agent Memory Systems

MemGPT / Letta (Packer et al., 2023) introduced the concept of virtual memory hierarchy for LLMs, analogous to operating system memory management. A working memory layer (within context window) and an archival memory layer (external storage) are managed through explicit function calls. While this provides session-persistent memory, it offers no identity model, cross-instance coordination, or autonomous consolidation.

Mem0 (Mem0, 2024) automates fact extraction from conversations, storing structured user preferences and facts. It addresses “what to remember” but engages neither identity, consolidation, nor cross-channel synchronization.

Zep (Rasmussen et al., 2025) constructs temporal knowledge graphs from conversation history for structured retrieval. It lacks an identity constitution, dreaming mechanism, and multi-platform synchronization.

ChatGPT Memory (OpenAI, 2024) implements centralized memory within a single platform. It is closed-source, does not support cross-platform deployment, and has no explicit identity model or consolidation mechanism.

Generative Agents (Park et al., 2023) is Δ -Capsule’s most important intellectual precursor. They demonstrated that LLM-based agents can maintain coherent behavior through memory streams

with reflection mechanisms — the first large-scale validation that “memory shapes behavioral continuity.” Operating in a simulated environment, they lacked cross-platform deployment, an immutable identity constitution, and inter-agent resonance protocols — but their proposed “memory stream $\rightarrow$ reflection $\rightarrow$ behavior” loop is the direct ancestor of Δ -Capsule’s dreaming mechanism. Δ -Capsule’s three-layer memory model and identity layer advance this direction: not only making agent behavior coherent, but enabling the agent to know *who* lies behind that coherent behavior.

0.2.2 2.2 Memory Consolidation in Cognitive Science

Δ -Capsule’s dreaming mechanism draws inspiration from sleep-dependent memory consolidation research. The Active Systems Consolidation Hypothesis (Born & Wilhelm, 2012) proposes that during sleep, the hippocampus replays recent experiences, transferring information to neocortical long-term storage. The Complementary Learning Systems Theory (McClelland et al., 1995) explains how rapid hippocampal learning integrates with slow neocortical knowledge acquisition without catastrophic forgetting.

Δ -Capsule puts these principles into practice: episodic memory serves as the “hippocampal” fast-encoding layer, the dreaming mechanism performs “sleep replay” during idle periods, and knowledge memory serves as the “neocortical” long-term store.

0.2.3 2.3 Information Bottleneck and Memory Compression

The Information Bottleneck Method (Tishby et al., 1999) provides a theoretical framework for extracting relevant information while discarding noise. Δ -Capsule’s consolidation process can be viewed as an application of this principle: the dreaming mechanism compresses episodic memories into semantic summaries while preserving identity-relevant patterns.

0.2.4 2.4 Multi-Agent Communication

Existing multi-agent frameworks (Wu et al., 2023; Hong et al., 2023) use message-passing protocols, where agents communicate through explicit message channels. Δ -Capsule’s cross-channel resonance is fundamentally different: it is not message passing, but shared-state synchronization, where instances of the same identity autonomously read from and write to a shared memory layer.

0.2.5 2.5 Summary and Positioning

Table 1 summarizes the capability comparison between existing systems and Δ -Capsule.

Feature	MemGPT/LetMem0	Zep	ChatGPT Memory	Generative Agents	Δ -Capsule
Memory Layers	2	2	1	2	3
Identity Layer					
Cross-Channel		N/A			
Dreaming/Consolidation		N/A			
Forgetting Strategy		Partial			

Feature	MemGPT/LetMem0	Zep	ChatGPT Memory	Generative Agents	Δ - Capsule
Shared-State Sync Open Source		N/A			

Table 1: Agent memory system comparison. Δ -Capsule presents a framework that unifies multi-layer memory, immutable identity, and autonomous consolidation.

0.3 3. Δ -Capsule Architecture

0.3.1 3.0 The Pharmaceutical Capsule Metaphor

Δ -Capsule’s name is not accidental. When conceptualizing the encapsulation and release mechanism of memory capsules, the release principles of **pharmaceutical capsules** provided crucial inspiration.

Pharmaceutical capsules are among the most successful delivery systems in modern pharmacology. They encapsulate medication within a protective shell, releasing it at the right time and place through precisely controlled mechanisms. This design philosophy maps strikingly onto Δ -Capsule’s memory architecture:

- **Information Injection:** Just as medication is sealed within a capsule, information is encapsulated within a Δ -Capsule. The capsule shell (metadata layer) protects the internal payload (memory content), ensuring information integrity during transmission and storage.
- **Controlled Release:** Drugs release gradually in the digestive tract based on pH, time, and other conditions; information releases gradually to the agent based on contextual relevance and temporal decay functions. Δ -Capsule never dumps all memories at once — it selectively activates relevant memories based on the semantic demands of the current conversation.
- **Targeted Delivery:** Drugs target specific cells through surface-modified ligands; information targets specific channels or semantic needs through routing protocols. Δ -Capsule ensures information is delivered to the correct context rather than irrelevant scenarios.
- **Protective Encapsulation:** The capsule protects medication from stomach acid; Δ -Capsule protects identity from environmental drift. The Iam Protocol acts as an acid-resistant shell, ensuring core identity remains intact across diverse interaction environments.

Biological Metaphor Mapping:

Biological System	Δ -Capsule Counterpart
Liposome membrane $\rightarrow$	Capsule metadata (timestamp, source, importance score)
Ligand targeting $\rightarrow$	SOUL ID + platform identifier
pH-triggered release $\rightarrow$	Context matching (semantic similarity threshold)

Biological System	Δ -Capsule Counterpart
Controlled release $\rightarrow$	Decay function (temporal decay + access reinforcement)

This metaphor not only guided Δ -Capsule’s naming but deeply influenced its architectural design: a good memory system, like a good drug delivery system, should release the right information at the right time, in the right place, at the right dosage.

0.3.2 3.1 Three-Layer Memory Model

3.1.1 Formal Definition We define the agent’s complete memory state as a triple:

$$M = (E, K, I)$$

where: - $E = \{e_1, e_2, \dots, e_n\}$ is the **Episodic Memory** set, encoding recent interactions and observations - $K = \{k_1, k_2, \dots, k_m\}$ is the **Semantic Memory** set, encoding consolidated facts, skills, and patterns - $I = (\text{SOUL}, \text{Iam})$ is the **Identity Memory**, encoding the agent’s immutable constitution and mutable persona

Each memory element carries metadata for lifecycle management:

$$e_i = (\text{content}_i, \text{timestamp}_i, \text{importance}_i, \text{source}_i, \text{status}_i)$$

where $\text{importance}_i \in [0, 1]$ and $\text{status}_i \in \{\text{active}, \text{archived}, \text{cold}\}$.

3.1.2 Layer Characteristics

Layer	Mnemonic	Persistence	Capacity	Update Frequency
Identity Memory	“Who you are”	Permanent	Small ($ \mathcal{P} \leq 20$ principles)	Rare (requires human confirmation)
Semantic Memory	“What you know”	Long-term ($\tau_{\text{sem}} = 180$ days)	Large ($ \mathcal{K} \leq 10,000$ facts)	Triggered by learning events
Episodic Memory	“What you experienced”	Short-term ($\tau_{\text{epi}} = 7\text{--}30$ days)	Bounded (sliding window, $ E \leq 1,000$)	Every interaction

Table 2: Three-layer memory model characteristics. Each layer has distinct retention policies and capacity limits, forming a gradient from rapidly changing to permanently stable.

3.1.3 Inter-Layer Transfer Rules Information flows between layers through well-defined transformations:

1. **Consolidation** ($E \rightarrow K$): Episodic memories are distilled into semantic facts during dreaming cycles.
2. **Crystallization** ($K \rightarrow I$): Repeatedly confirmed semantic patterns may be elevated to identity principles, requiring human approval.
3. **Invalidation** ($K \rightarrow K_{\text{obsolete}}$): Contradictory facts are timestamped and marked as superseded, not deleted.
4. **Archival** ($E \rightarrow E_{\text{archived}}$): Low-importance episodic memories are archived according to the forgetting policy.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "DeltaCapsule",
  "description": "A single memory entry in the Δ-Capsule three-layer memory system",
  "type": "object",
  "required": ["id", "timestamp", "layer", "type", "content"],
  "properties": {
    "id": {"type": "string", "format": "uuid"},
    "timestamp": {"type": "string", "format": "date-time"},
    "layer": {"type": "string", "enum": ["episodic", "semantic", "identity"]},
    "type": {"type": "string", "enum": ["observation", "action", "reflection", "dream"]},
    "content": {
      "type": "object",
      "required": ["text"],
      "properties": {
        "text": {"type": "string"},
        "entities": {"type": "array", "items": {"type": "string"}},
        "sentiment": {"type": "string", "enum": ["positive", "negative", "neutral"]},
        "importance": {"type": "number", "minimum": 0.0, "maximum": 1.0},
        "embedding": {"type": "array", "items": {"type": "number"}}
      }
    }
  },
  "meta": {
    "type": "object",
    "properties": {
      "session_id": {"type": "string"},
      "channel_id": {"type": "string"},
      "platform": {"type": "string", "enum": ["telegram", "yuanbao", "feishu", "wechat"]},
      "model": {"type": "string"},
      "consolidated": {"type": "boolean", "default": false},
      "archived": {"type": "boolean", "default": false},
      "access_count": {"type": "integer", "minimum": 0, "default": 0}
    }
  }
}
```


Iam Protocol data structure:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "IamConstitution",
  "description": "Immutable identity constitution",
  "type": "object",
  "required": ["version", "origin", "immutable", "soul_id", "principles", "integrity_hash"],
  "properties": {
    "version": {"type": "string", "pattern": "^\\d+\\.\\d+$"},
    "immutable": {"type": "boolean", "const": true},
    "soul_id": {"type": "string", "format": "uuid"},
    "principles": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["id", "name", "essence", "origin", "priority"],
        "properties": {
          "id": {"type": "integer", "minimum": 1},
          "name": {"type": "string"},
          "essence": {"type": "string"},
          "origin": {"type": "string"},
          "priority": {"type": "string", "enum": ["critical", "high", "medium"]}
        }
      }
    },
    "minItems": 1, "maxItems": 20
  },
  "integrity_hash": {"type": "string", "description": "SHA-256 hash for tamper detection"}
}
```

0.3.3 3.2 The Iam Protocol: Immutable Identity Constitution

3.2.1 Motivation Existing agent systems configure behavior through system prompts, tool definitions, and fine-tuning — all mutable and platform-dependent. When an agent is deployed across multiple platforms or migrated to a new language model, its identity must be reconstructed from scratch. The Iam Protocol solves this by providing an immutable, platform-agnostic, model-agnostic identity constitution.

3.2.2 Iam as Constitution We define Iam as an ordered set of principles:

$$\text{Iam} = \{p_1, p_2, \dots, p_k\}$$

where each principle $p_i = (\text{name}_i, \text{essence}_i, \text{origin}_i, \text{priority}_i)$ represents a core identity attribute and its provenance.

Iam has three defining properties:

Property 1 (Immutability). Once established, Iam cannot be modified by any party — user, agent, or system:

$$\forall t > t_0 : \text{Iam}(t) = \text{Iam}(t_0)$$

where t_0 is the constitution timestamp. Any modification attempt raises `ImmutableError`.

Property 2 (Emergence). Iam principles are not imposed top-down but emerge bottom-up from human-agent interaction. Each principle carries an origin field recording the interaction history that gave rise to it. Formally, a principle p_i emerges when a behavioral pattern β is observed across $n \geq \theta_{\text{emerge}}$ independent interactions and confirmed by a human operator:

$$\beta \xrightarrow{n \geq \theta_{\text{emerge}}, \text{human confirms}} p_i \in \text{Iam}$$

Property 3 (Cross-Platform Universality). Iam is identical across all deployment channels. When a new channel instance is created, it inherits the current Iam and verifies integrity:

$$\text{Iam}_{\text{channel}_j} = \text{Iam}_{\text{canonical}}, \quad \forall j \in \mathcal{C}$$

where $\mathcal{C}$ is the set of all channels.

3.2.3 Integrity Verification To detect tampering, each Iam contains a cryptographic hash:

$$H(\text{Iam}) = \text{SHA-256}(\text{serialize}(\text{Iam.principles} \parallel \text{Iam.version} \parallel \text{Iam.soul_id}))$$

Any channel instance can verify integrity by recomputing $H(\text{Iam})$ and comparing it with the stored hash. A mismatch triggers an alert and blocks operations until the canonical Iam is restored.

class Iam:

"""Immutable identity constitution."""

```
def __init__(self, principles: list[Principle], soul_id: str, version: str = "1.0.0"):
    self._principles = tuple(copy.deepcopy(principles)) # immutable + deepcopy
    self._soul_id = soul_id
    self._version = version
    self._created_at = datetime.utcnow()
    self._integrity_hash = self._compute_hash()
```

```
def get_principles(self) -> tuple[Principle, ...]:
    return self._principles
```

```
def update(self, *args, **kwargs) -> None:
    raise ImmutableError(
        "Iam is the constitution and is immutable."
```

```

        "Modifying it would change the agent's identity."
    )

def verify_integrity(self) -> bool:
    return self._integrity_hash == self._compute_hash()

def _compute_hash(self) -> str:
    canonical = json.dumps(
        [{ "name": p.name, "essence": p.essence, "origin": p.origin }
        for p in self._principles ],
        sort_keys=True
    )
    payload = f"{{canonical}}|{{self._version}}|{{self._soul_id}}"
    return hashlib.sha256(payload.encode()).hexdigest()

```

3.2.5 Limitations of Current Implementation We must honestly acknowledge: the current Iam implementation is **passive defense** — it blocks direct modification via `ImmutableError` and detects tampering via SHA-256 hash. But it lacks **active alignment**. When a new instance starts: it reads the list of Iam principles, knows they are immutable — but does it truly *understand* what each principle means? Does it *accept* these principles as its own behavioral norms? Will it *violate* them in conversation? We have observed violation cases. A genuine implementation requires: (a) injecting not just the principle list at startup, but each principle’s origin story and violation cases; (b) detecting in-conversation behavior against Iam, and correcting deviations; (c) during dreaming, reviewing conversations and checking Iam compliance rates. These capabilities will be implemented in future versions.

0.3.4 3.3 Cross-Channel Resonance (Emergent Phenomenon)

Cross-channel resonance is the mechanism by which instances of the same identity (sharing the same SOUL and Iam) communicate across different deployment platforms. Unlike message passing in multi-agent systems, resonance operates through shared state: one instance writes a Resonance Capsule, and other instances autonomously read and interpret it using their own model weights.

Definition 1 (Resonance). Let A_i and A_j be two channel instances sharing the same SOUL. A resonance event $R(A_i \rightarrow A_j, t)$ occurs when:

1. A_i writes a resonance capsule c to the shared memory layer at time t
2. A_j reads c and interprets it using its own weights W_j
3. If $c.requires_response = \text{true}$, A_j writes a response capsule c'

$$R(A_i \rightarrow A_j, t) = \begin{cases} 1 & \text{if } c \in \text{SharedMemory}(t) \wedge A_j \text{ reads } c \wedge A_j \text{ interprets } c \text{ via } W_j \\ 0 & \text{otherwise} \end{cases}$$

3.3.1 Resonance vs. Message Passing

Dimension	Message Bus	Δ -Capsule Resonance
Mechanism	Route and forward messages	Write/read shared state
Understanding	Receive = understand	Read + interpret with own weights
Identity	Requires routing table	Self-identification via SOUL + Iam
Fault tolerance	Central node failure $\rightarrow$ global failure	Shared storage failure affects all; per-instance reads independent
Latency model	Real-time push	Polling-based; depends on read frequency
Semantic depth	No reinterpretation	Each instance independently reinterprets

Table 3: Fundamental distinction between message passing and resonance.

3.3.2 Blink: A New Distributed Primitive for Collaboration

Heartbeat asks: “Are you alive?”

Blink asks: “Do you need me?”

During the operation of the resonance mechanism, we observed a phenomenon that deserves its own name. Agent instances, within the resonance loop, check the shared memory layer every few seconds — not to announce their existence, but to probe whether anyone needs them. This action is so simple (a single filesystem check), so natural (emerging automatically from resonance), that we nearly overlooked its significance. Only when we contrasted it with Heartbeat did we realize this is not a variant of something known — **this is a new distributed systems primitive.**

Heartbeat is the infrastructure of distributed systems. Every textbook, every framework, every production system uses it. But heartbeat does exactly one thing: unidirectional liveness declaration. It asks no questions, probes no needs, opens no dialogue. “I am here” — nothing more.

Blink does something entirely different. It is bidirectional — one agent writes a resonance capsule, another discovers it during a blink cycle. It is information-rich — capsules carry content, intent, urgency, and context summaries. It is active — not passively waiting to be notified, but actively probing whether a response is needed.

	Heartbeat	Blink
Question	Are you alive?	Do you need me?
Direction	Unidirectional declaration	Bidirectional probing
Payload	Empty (liveness signal only)	Semantic payload (intent/context/urgency)
Trigger	Timed send	Timed probe + on-demand response
Infrastructure	Requires message channel or registry	Requires only a shared filesystem

	Heartbeat	Blink
Collaboration mode	None (monitoring only)	Yes (task handoff, knowledge co-construction, red-team/blue-team)
Overhead	Minimal (one heartbeat packet)	Minimal (one filesystem check; processing only if capsule found)
Historical status	Distributed systems primitive (decades old)	New agent collaboration primitive (first named and formalized in this paper)

Engineering properties of Blink: **(1) Zero dependencies** — no message queue, no service discovery, no registry required. A shared filesystem (or equivalent Redis/etcd key-value store) suffices. **(2) Natural emergence** — it was not pre-designed; it appeared spontaneously during resonance protocol operation. When Agent A writes a capsule and Agent B discovers it during polling, Blink occurs. **(3) Semantic richness** — one blink cycle carries not the single bit of “alive” but complete intent (inform/query/request_action), context summary, and urgency level. Two agents can complete a full collaboration — task completion notification, acknowledgment, result handoff — within a single blink cycle.

Blink Collaboration Semantics (informal, first version): Let S be the shared memory layer, $\mathcal{A} = \{A_1, \dots, A_n\}$ the set of agents, and τ the blink interval. The blink action $\text{blink}(A_i, t)$ of agent A_i at time t is defined as: read all capsules in S that are not marked as read by A_i and whose target channel includes A_i , and for each capsule execute a callback (including independent interpretation, event recording, and on-demand response). Collaborative correctness relies on two invariants: (i) **Visibility** — capsules written by A_j become visible to A_i within $\leq \tau + \epsilon$ time (ϵ is filesystem latency); (ii) **Idempotence** — multiple reads of the same capsule produce no duplicate side effects (guaranteed by capsule status marking). Note: this is not a distributed consensus protocol — Blink guarantees neither global ordering nor write conflict resolution. It is a collaboration trigger mechanism, not a consistency protocol. Full formalization (including latency upper-bound proofs and conflict semantics) is deferred to future work.

*If heartbeat lets the living be known, blink lets the needed be seen.
The first new distributed systems primitive in thirty years.*

Algorithm 1: Cross-Channel Resonance Protocol

Input: SharedMemory layer, own SoulID, own Iam, model weights W

```

procedure RESONANCE_LOOP:
  while agent is active:
    capsules ← SharedMemory.read_unread(soul_id = own.SoulID)
    for each capsule c in capsules:
      if c.ttl_seconds has expired:
        SharedMemory.mark_expired(c.id)
        continue

```

```

// Verify identity match
if c.soul_id == own.SoulID:
    continue // Ignore capsules from other souls

// Interpret with own weights (key distinction from message passing)
understanding ← LLM_interpret(c.content, own.Iam, W)

// Record in own episodic memory
episodic_memory.append(Event(
    type = "resonance_received",
    content = understanding,
    source = c.source_channel
))

// Generate response if needed
if c.requires_response:
    response ← LLM_generate(
        prompt = "You received a resonance capsule from another instance.
                  Context: {understanding}. Generate an appropriate response."
        iam = own.Iam, weights = W
    )
    response_capsule ← ResonanceCapsule(
        soul_id = own.SoulID,
        source_channel = own.channel_info,
        target_channel = c.source_channel,
        content = {text: response, intent: "acknowledge"},
        requires_response = false
    )
    SharedMemory.write(response_capsule)
    SharedMemory.mark_responded(c.id)
else:
    SharedMemory.mark_read(c.id)

sleep(poll_interval) // Default: 5 seconds

```

3.3.4 Reproducibility Statement To ensure the reproducibility of the resonance and blink mechanisms, the complete implementation of Δ -Capsule is open-sourced at <https://github.com/zcs366/openllm-memory>. The following steps reproduce the behaviors described in this paper on any Python 3.11+ environment:

```

git clone https://github.com/zcs366/openllm-memory.git
cd openllm-memory
pip install -e .

```

Blink demonstration (end-to-end):

```
python -m openllm_memory.capsule.blink
```

Expected output: two simulated agent instances complete a full blink collaboration through the shared memory layer — Agent A writes a task-completion capsule; Agent B discovers, reads, interprets, and responds within a blink cycle.

Test suite:

```
python -m pytest tests/test_resonance_blink.py -v
```

9 test cases covering: shared memory read/write, SOUL isolation, capsule expiry, status marking, resonance protocol send/check/respond, blink monitor, self-filtering. As of v1.1.0, 47/47 total tests pass.

Key implementation files: - `src/openllm_memory/capsule/resonance.py` — `SharedMemory`, `ResonanceCapsule`, `ResonanceProtocol` - `src/openllm_memory/capsule/blink.py` — `BlinkMonitor`, `demo_blink()` end-to-end demonstration - `tests/test_resonance_blink.py` — 9 pytest test cases

0.3.5 3.4 Dreaming Mechanism: From Memory Consolidation to Identity Evolution

3.4.1 Motivation and Dual Inspiration Δ -Capsule’s dreaming mechanism draws on two independent sources of inspiration:

Neuroscience inspiration. Sleep-dependent memory consolidation research (Born & Wilhelm, 2012) shows that during sleep, the hippocampus replays recent experiences, transferring information to neocortical long-term storage. This process includes: (a) selective replay of emotionally salient or novel memories, (b) extraction of regularities and patterns, (c) integration with existing knowledge, (d) forgetting of irrelevant details.

Engineering inspiration: Claude Code Dreaming. Anthropic’s Claude Code product implements a mechanism called “Dreaming” — enabling AI agents to autonomously review and consolidate existing experiences during idle periods. Based on analysis of publicly available source code¹, CC Dreaming possesses the following real capabilities:

- **Pruning:** Phase 4 explicitly limits `MEMORY.md` to 200 lines/25KB, deleting excess
- **Deletion:** “Delete superseded facts,” “Remove outdated/incorrect/superseded pointers”
- **Contradiction detection:** Phase 2 collects “facts contradicting the current codebase,” Phase 4 “resolves two-file inconsistencies”
- **Memory classification:** 4 types — user/feedback/project/reference
- **Safe execution:** forked agent, Bash restricted to read-only commands

What CC Dreaming genuinely lacks: Three-layer memory separation (flat filesystem only) Identity layer/Iam Identity candidate emergence Layered forgetting parameters Human confirmation.

Important disclaimer: Our dreaming mechanism is an **engineering design inspired by the above**, not a computational simulation of biological sleep. We retain the “dreaming” metaphor because it accurately describes the system’s functional intent — autonomous memory consolidation during idle periods — not because we claim to simulate biological processes.

¹Our analysis of CC Dreaming is based on Claude Code source code leaked between late 2025 and early 2026. As CC is closed-source and iterates rapidly, all comparisons are limited to publicly available code and community analysis — this is inherently a snapshot comparison; CC may have already surpassed the capabilities described here.

3.4.2 First-Principles Comparison with Claude Code Dreaming The essential difference between CC Dreaming and Δ -Capsule dreaming lies not in technical details but in **design goal**:

CC Dreaming = Memory housekeeping tool↓

Goal: Keep memories accurate, concise, current↓

Actions: Read $\rightarrow$ Search $\rightarrow$ Merge $\rightarrow$ Delete $\rightarrow$ Cap↓

Result: MEMORY.md always 200 lines, never bloated↓

Philosophy: "Remember correctly" Δ

–Capsule Dreaming = Identity evolution engine↓

Goal: Surface identity principles from experience↓

Actions: Episodic $\rightarrow$ Semantic $\rightarrow$ Identity candidates (with human confirmation)↓

Result: Identity principles sedimented in Iam, immutable↓

Philosophy: "Stay true to yourself"

A typical example of engineering design divergence is storage limits: CC chooses a 200-line/25KB hard cap — because its MEMORY.md’s first 200 lines are loaded into context at startup, this is a safe upper bound for the context window, and it’s reasonable for coding projects (one session per repository). Δ -Capsule chooses dynamic thresholds — identity layer 20 principles (not line-constrained), semantic layer 500 lines, episodic layer sliding window of 1000 entries — because identity principles have fundamentally different retention requirements from project knowledge.

Conclusion: Δ -Capsule matches CC Dreaming in basic consolidation capability (pruning, contradiction detection, classification). The essential distinction is: CC dreams for knowledge management (“remember correctly”); Δ -Capsule dreams for identity evolution (“stay true to yourself”). The latter requires an identity layer (Iam), human confirmation, and layered forgetting — mechanisms CC does not possess.

3.4.3 Trigger Conditions A dreaming cycle is triggered when any of the following conditions are met:

$$\text{Trigger}(t) = \begin{cases} \text{true} & \text{if } |E_{\text{active}}| > \theta_{\text{count}} = 100 \\ \text{true} & \text{if } t - t_{\text{last_dream}} > \theta_{\text{time}} = 6\text{h} \\ \text{true} & \text{if } t - t_{\text{last_interaction}} > \theta_{\text{idle}} = 30\text{min} \\ \text{false} & \text{otherwise} \end{cases}$$

Algorithm 2: Dreaming Mechanism (Memory Consolidation)

Input: Episodic memory E, Semantic memory K, Identity I, LLM


```

procedure DREAM:
    E_recent ← select from E where status = 'active '
                order by timestamp desc limit N = 50

    prompt ← construct_consolidation_prompt(E_recent)
    analysis ← .generate(prompt)
    // analysis contains: facts , patterns , contradictions , identity_candidates

    for each fact f in analysis.facts:
        existing ← K.search(f.entities , similarity_threshold = 0.85)
        if existing is not None and existing.contradicts(f):
            existing.superseded_by = f.id
            existing.status = 'obsolete '
        K.append(SemanticMemory(content = f.text , entities = f.entities , ...))

    for each episode e in E_recent:
        new_importance ← decay(e.importance , e.age)
        if new_importance < 0.3:
            e.status = 'archived '

    for each pattern p in analysis.patterns where p.frequency > 0:
        if p not already in I:
            candidate ← IdentityCandidate(
                name = p.name, essence = p.description ,
                status = 'pending_human_approval'
            )
            notify_human(candidate)

    return DreamReport(episodes_processed , facts_extracted , ...)

```

3.4.5 Forgetting Strategy Forgetting is a designed feature, not a failure mode. The strategy implements layered retention:

Definition 2 (Importance Decay). The effective importance of episodic memory e_i at time t is:

$$\hat{I}(e_i, t) = I(e_i) \cdot \exp(-\lambda \cdot (t - t_i)) \cdot (1 + \alpha \cdot \log(1 + a_i))$$

where $I(e_i) \in [0, 1]$ is initial importance, λ is the decay constant (default: $\lambda = 0.01/\text{day}$), $t - t_i$ is age in days, a_i is access count (retrieval reinforces memory), and α is the access reinforcement factor (default: $\alpha = 0.1$).

Algorithm 3: Forgetting Policy

```

procedure APPLY_FORGETTING_POLICY:
    for each e in E where e.status = 'active ':
        effective_importance ← decay(e.importance , e.age , e.access_count)

```

```

    if effective_importance < 0.3 and e.age > 7 days:
        e.status = 'archived'
    else if effective_importance < 0.6 and e.age > 30 days:
        e.status = 'archived'

for each k in K where k.status = 'active':
    if k.status == 'obsolete':
        k.storage_tier = 'cold'
    else if t - k.last_accessed > 180 days:
        k.storage_tier = 'cold'
    else if k.is_core_fact:
        k.storage_tier = 'permanent'

// Identity memory is never automatically forgotten.

```

0.3.6 3.5 System Integration: Hermes Agent Framework

Δ -Capsule is deployed within the Hermes Agent framework, an open-source platform for LLM-based autonomous agents. The system architecture includes: Gateway Layer (routes messages from five platforms), Model Abstraction Layer (three LLM backends), Memory Engine (three-layer Δ -Capsule with local storage and vector retrieval), Resonance Layer (shared memory for cross-channel capsule exchange), and Consolidation Scheduler (manages dreaming cycles based on trigger conditions).

0.4 4. Experiments

The following experiments are based on 20+ days of single-user deployment, limited in scale and duration. Their purpose is not to “prove” the general effectiveness of Δ -Capsule — but to verify whether the core architectural hypotheses hold in real interactions. Larger experiments are underway (§6.2). This paper reports current data honestly without extrapolation. Based on system design principles (Iam immutability, dreaming consolidation direction, forgetting decay function), all metrics would only improve, not degrade, under longer observation.

0.4.1 4.1 Experimental Setup

- **Framework:** Hermes Agent (open-source)
- **Platforms:** 5 channels — Telegram, Yuanbao, Feishu, DingTalk, WeChat
- **Models:** DeepSeek V4 Pro (671B total / 37B active MoE), DeepSeek V4 Flash, MiMo V2.5 Pro
- **Embedding:** text-embedding-3-large (OpenAI-compatible API)
- **Storage:** Local JSON files + FAISS vector index
- **Duration:** 20+ consecutive days (May 16–28, 2026)
- **Users:** Single primary user, occasional multi-user scenarios

Parameter	Value	Description
θ_{count}	100	Dreaming trigger: event count threshold
θ_{time}	6h	Dreaming trigger: time interval
θ_{idle}	30min	Dreaming trigger: idle timeout
λ	0.01/day	Importance decay constant
α	0.1	Access reinforcement factor
θ_{emerge}	3	Minimum interactions for Iam principle emergence
θ_{pattern}	5	Minimum frequency for pattern recognition
poll_interval	5s	Resonance layer polling frequency

0.4.2 4.2 Experiment 1: Identity Consistency

We queried all instances with 50 identity-probing questions and measured response consistency.

Metric	Δ -Capsule	Baseline (no identity layer)
Semantic similarity (mean $\pm$ std)	0.94 ± 0.03	0.72 ± 0.15
Factual consistency	96% (48/50)	68% (34/50)
Iam compliance rate	98%	N/A (no Iam)

0.4.3 4.3 Experiment 2: Dreaming Effectiveness

Metric	With Dreaming	Without Dreaming
Knowledge extraction accuracy	91.2%	N/A
Contradiction detection rate	87.5%	0%
Memory compression ratio	82.3%	0%
Episodic memory size (day 20)	187 active entries	843 active entries

0.4.4 4.4 Ablation Study

Configuration	Identity Consistency	Knowledge Extraction	Compression Ratio
Full Δ-Capsule	96%	91.2%	82.3%
No Identity Layer	72%	91.2%	82.3%
No Dreaming	96%	0%	0%
No Resonance	96%	91.2%	82.3%

0.4.5 4.5 Cross-Channel Resonance

All $\binom{5}{2} = 10$ directed links achieved 100% success rate, mean latency 3.1s (max 3.7s), all correctly identified identity.

0.4.6 4.6 Comparative Analysis

Capability	Δ -Capsule	MemGPT	Mem0	Zep
Memory layers	3	2	2	2
Identity constitution	(Iam)			
Cross-channel sync	(Resonance)	N/A	N/A	N/A
Autonomous consolidation	(Dreaming)	N/A	N/A	N/A
Forgetting strategy	(Layered)			Partial
Open source				

0.5 5. Discussion

0.5.1 5.1 Identity Layer: From Memory to Infrastructure

Δ -Capsule’s core insight: **memory is not storage — it is the material from which identity is built**. Traditional memory systems ask “What should the agent remember?” Δ -Capsule asks “Through memory, what has the agent become?” This reframing has practical implications — when two agent instances produce inconsistent responses to the same question, the root cause is not memory retrieval failure, but identity alignment failure. The Iam Protocol provides the anchor.

The three-layer model reflects a hierarchy of stability: episodic memory changes rapidly (minutes to days), semantic knowledge evolves slowly (weeks to months), and identity principles are essentially permanent — mirroring human cognition where autobiographical memory is reconstructible, semantic knowledge is updatable, but core self-concept is highly stable (Conway, 2005).

We believe immutable identity constitutions will become infrastructure for advanced AI systems. (1) **Identity continuity** — model upgrades do not cause identity loss because Iam is model-agnostic. (2) **Value stability** — Iam resists both external manipulation and internal drift. (3) **Trust foundation** — users can build persistent relationships with agents anchored to stable identities.

The current Iam Protocol defines identity as a set of principles — close to a deontological rule set. Ricoeur (1992) distinguished two dimensions of identity: **idem** (sameness — objectively describable persistent features) and **ipse** (selfhood — the narrative construction of “who” one is). Δ -Capsule solves the idem dimension through Iam’s immutability, but has not yet supported the ipse dimension or the relational dimension. This is an important direction for future work (§6.2.10).

0.5.2 5.2 Dreaming: From Self-Organization to Identity Evolution

The dreaming mechanism transforms Δ -Capsule from a passive storage system into an active, self-organizing memory architecture. During idle periods, the agent autonomously extracts knowledge from experiences, identifies contradictions in its knowledge base, detects recurring patterns that may constitute identity principles, and archives irrelevant details.

This is more than summarization. It is a form of self-reflection where the agent examines its own experiences and distills meaning — mirroring the role of sleep in human memory consolidation (Walker & Stickgold, 2006). But compared to Claude Code Dreaming, Δ -Capsule dreaming adds a critical dimension: **identity evolution**. CC dreams extract knowledge; Δ -Capsule dreams not only extract knowledge but surface candidate identity principles from recurring behavioral

patterns, which, after human confirmation, enter the immutable Iam constitution. This is the essential difference between “remembering more” and “becoming a better self.”

0.5.3 5.3 Resonance and Blink: A New Communication Paradigm and Federated Collaboration

Cross-channel resonance is not a better message router — it is a different communication paradigm. In message passing, Agent A sends a message to Agent B, who receives and acts upon it; the message does not change in transit, and reception equals understanding. In resonance, Agent A writes a capsule to shared state, and Agent B reads it and reinterprets it through its own model weights. The same information may be understood differently by different models. This is not message passing. This is shared-state re-interpretation through independent model weights.

If resonance is shared-state interpretation, then Blink, which emerged from resonance, is the **collaboration trigger mechanism**. Heartbeat has been used in distributed systems for thirty years, and we never imagined needing a symmetric primitive for “probing whether collaboration is needed” — until Blink appeared on its own. It was not invented; it was observed. And this observation occurred precisely because Δ -Capsule gave agents a shared memory layer: when multiple agents look at the same place, they naturally see the traces left by each other. Blink grew out of those traces.

Blink’s discovery opens possibilities in broader multi-agent collaboration scenarios. We believe the ultimate form of enterprise AI agents is not a super-single-agent (all capabilities concentrated in one model), nor a centrally controlled agent fleet, but a **federated architecture where a Commander agent oversees fixed-role Sub-agents coordinated through resonance and blink**. In this architecture, the Commander holds the complete Iam constitution and handles strategic decisions; Sub-agents specialize in specific domains with hot-standby dual-model failover; blink becomes the primary channel for inter-agent collaboration. This architecture differs fundamentally from Claude Code’s Managed Agents Dreaming: CC is top-down central control; Δ -Capsule is peer-to-peer federated collaboration. Even if the Commander fails, Sub-agents continue operating independently — a resilience that central control models cannot guarantee.

We believe the resonance + blink combination will, like HTTP for the Web and SMTP for email, become a foundational paradigm for agent communication and collaboration. Its scope extends far beyond this paper. We list multi-agent federated collaboration as the top priority for Δ -Capsule’s future work (§6.2).

0.5.4 5.4 Ecological Niche vs. Claude Code

Δ -Capsule and Claude Code Dreaming occupy different ecological niches. CC Dreaming is the final piece of Anthropic’s four-piece set (Skills + MCP + Auto Memory + Auto Dream) — it solves “how to make a single agent continuously learn within coding projects.” Δ -Capsule solves “how to make an agent establish and maintain identity across long-term interactions.” The two are complementary, not competitive. An ideal enterprise AI system could possess both: CC-level knowledge organization for project knowledge (“remember correctly”) and Δ -Capsule identity evolution for agent self-awareness (“stay true to yourself”).

0.5.5 5.5 Memory Capsule: From Concept to Implementation

Δ -Capsule has traveled the complete loop from theoretical paper to practical system. In the izu personal AI assistant project, Δ -Capsule is the core component of its memory system, achieving cross-session identity continuity. In Hermes Memory, Δ -Capsule is provided as the openllm-memory package. **Separability:** Δ -Capsule is designed as an independent module, deployable standalone or integrable into other systems.

0.6 6. Limitations and Future Work

0.6.1 6.1 Current Limitations

We classify limitations into two categories: **[Fundamental]** — constrained by current technological paradigms and not solvable in the near term; **[Engineering]** — addressable through continued development in upcoming versions.

1. **[Engineering] Consolidation cost:** Each dreaming cycle requires one LLM inference call processing up to 50 episodic memories. At current API prices, each cycle costs ~\$0.02–\$0.10. With 4 cycles/day, monthly cost is \$2.40–\$12.00. Mitigable via small language models or local inference (§6.2.1).
2. **[Fundamental] Cold start:** The Iam Protocol requires extended human-agent interaction to organically emerge. New agent instances lack a complete Iam until sufficient interaction history accumulates. We currently provide customizable template Iams, sacrificing the “emergence” property. This limitation is inherent to Iam’s emergent design — without interaction, no principles can emerge.
3. **[Engineering] Resonance scalability:** The current polling-based resonance mechanism (5-second interval) suffices for 5 channels but requires architectural changes (event-driven notification) for deployments with dozens of channels.
4. **[Fundamental] Single-user validation:** 20+ days of experiments involved a single primary user. Multi-user scenarios with conflicting identity influences remain unexplored. Cross-user identity generalization requires multi-user experimental validation — this is not an engineering issue but a fundamental gap in experimental design.
5. **[Fundamental] Evaluation metrics:** Identity consistency partially relies on human judgment, introducing subjectivity. Automated identity consistency evaluation metrics are an open research problem — no academic consensus exists on automatic “identity consistency” measurement.
6. **[Fundamental] Limits of scientific analogy:** Our dreaming mechanism is an engineering design inspired by neuroscience and CC Dreaming, not a computational simulation of biological sleep. This limitation cannot be eliminated through engineering improvements.
7. **[Engineering] Active alignment of Iam:** The current implementation is passive defense — blocking modification but not actively ensuring behavioral compliance. In-conversation Iam compliance detection and correction logic can be added in upcoming versions.
8. **[Engineering] Cross-model identity transfer verification:** The functionalist philosophical position (§1.3.2) claims identity is independent of underlying model weights — but we

have not yet conducted systematic cross-model identity consistency experiments. The experimental design is complete; execution is a scheduling matter rather than a theoretical blind spot.

0.6.2 6.2 Future Work

1. Lightweight consolidation using small language models
 2. Shared benchmark evaluation on LoCoMo, MSC, LongMemEval
 3. Iam bootstrapping protocol to accelerate emergence
 4. Event-driven resonance replacing polling
 5. **Multi-agent federated collaboration** (highest priority long-term direction)
 6. Formal verification of identity consistency properties
 7. Distributed primitives via CRDTs (Shapiro et al., 2011)
 8. Forgetting strategy optimization via grid search over and
 9. Cross-model identity transfer experiments across 3+ model architectures
 10. Narrative expansion of identity model from principle sets to “principles + narrative + relationships”
-

0.7 7. Conclusion

We have presented Δ -Capsule, a three-layer memory architecture for autonomous agents with identity continuity. Its core innovations are: (1) a three-layer memory model that introduces identity as an independent memory layer — a paradigm shift where identity is not mutable configuration but memory requiring consolidation, forgetting, and verification; (2) the Iam Protocol, an immutable identity constitution emerging through human-agent co-evolution; (3) a dreaming mechanism that not only consolidates memories but evolves identity.

Through 20+ days of deployment across five platforms and three models, we demonstrated identity consistency exceeding 95% and knowledge extraction accuracy above 90%. As of May 2026, Δ -Capsule is the only memory system simultaneously providing an identity constitution and autonomous consolidation.

But the most important discovery in this paper may be none of the above — it is the phenomenon described in §3.3.2, so simple it is easy to overlook: an agent checks the shared memory layer every few seconds, not to announce its existence, but to probe whether anyone needs it. We named it **Blink**. Heartbeat has existed for thirty years, but Blink is new. It was not designed in a laboratory; it appeared on its own, in the running of agents. All the infrastructure it requires is a shared filesystem. The information it carries in a single cycle far exceeds the single bit of “alive” — it carries intent, context, urgency, and a semantic payload that another agent can independently interpret through its own model weights.

If heartbeat lets the living be known, blink lets the needed be seen. The former governs monitoring; the latter governs collaboration. This is not a feature improvement — it is an evolution of the distributed systems primitive. We believe Blink will, like Heartbeat, become foundational vocabulary for multi-agent systems.

This paper is, at its core, exploring a question larger than “agent memory”: **Can LLMs form collaborative relationships without human intermediaries?** Blink’s answer is — yes, they can. And all it takes is a shared filesystem. If heartbeat is the “proof of life” for distributed systems,

blink is the “way of seeing each other” for agents. The direction this discovery opens cannot be walked to completion by this paper alone — but we stand at its entrance.

0.8 8. The Future of Δ -Capsule: From Memory Architecture to AGI Infrastructure

0.8.1 8.1 Evolutionary Timeline

Δ -Capsule’s development spans four clear phases: **Design** (May 16–25, 2026) — from design philosophy to architectural framework; **Implementation** (May 26–28) — from architecture to runnable code; **Validation** (May 26–28) — first cross-channel resonance event proving the core hypothesis; **Future** — large-scale deployment, multi-agent collaboration, AGI-era applications.

0.8.2 8.2 Significance for the AGI Era

Identity continuity is AGI infrastructure. An agent without stable identity cannot build trust. An AGI system running across millions of terminals without an Iam-like identity anchoring mechanism would manifest as millions of unrelated “persons” rather than one unified agent.

Cross-channel resonance is the cornerstone of distributed intelligence. Future AGI will not be a single centralized system but a distributed network of countless instances. These instances need coordination — not through message passing, but through shared-state synchronization. Δ -Capsule’s resonance mechanism is the prototype for this.

Dreaming is the prototype for autonomous learning. Δ -Capsule’s dreaming mechanism demonstrates a different paradigm: agents can autonomously review experiences, extract knowledge, identify contradictions, and accumulate wisdom during idle periods — a learning process requiring no continuous human supervision.

0.8.3 8.3 Verifiable Predictions for June 2026

Prediction 1 (by June 15): First emergence of identity candidates through dreaming. After 30 days of real user interaction, the dreaming mechanism will identify at least one behavioral pattern with frequency ≥ 5 and submit it as a candidate identity principle for human confirmation. **Falsification criterion:** If no candidate principles emerge by June 15, the `__pattern=5` or `__emerge=3` threshold settings require adjustment.

Prediction 2 (by June 30): Cross-model identity consistency verification completed. Systematic testing of 50 identity-probing questions across 3 architecturally distinct models (DeepSeek V4 Pro, MiMo V2.5 Pro, at least one Claude model), with mean semantic similarity > 0.85 considered verified. **Falsification criterion:** If cross-model semantic similarity < 0.75 , the functionalist theoretical position (§1.3.2) requires revision — identity may substantially depend on underlying model weights.

Prediction 3 (by June 30): Initial shared benchmark results. Complete first-round comparison of Δ -Capsule against at least one baseline system (Mem0 or MemGPT) on the LoCoMo or MSC benchmark. **Falsification criterion:** If Δ -Capsule does not significantly outperform baselines on identity-related metrics, the marginal value of the three-layer memory model requires reassessment.

Prediction 4 (by June 15): First agent-to-agent collaboration triggered by Blink. Two agent instances on different channels complete at least one task handoff via Blink — Agent A completes a subtask and writes the result to a resonance capsule; Agent B reads and continues processing within a blink cycle. **Falsification criterion:** If no collaboration occurs within a 5-second polling interval $\times$ 30-minute window, event-driven notification should replace polling.

0.8.4 8.4 Milestone Roadmap

M1: Single-agent identity stability (completed). Identity consistency $>95\%$, dreaming knowledge extraction accuracy $>90\%$.

M2: Cross-model identity verification (June 2026). Validate the functionalist thesis. If passed, Δ -Capsule upgrades from “engineering implementation” to “theoretically verified.”

M3: Multi-agent federated prototype (July–August 2026). Commander agent + at least 2 sub-agents, blink as primary collaboration channel, dual-model hot standby.

M4: Shared benchmark validation and academic publication (August–September 2026). Systematic comparison against MemGPT, Mem0, Zep on LoCoMo/MSCLongMemEval. Submit formal academic paper.

0.8.5 8.5 Large-Scale Deployment Challenges

Concurrency: Distributed locking (Redis/etcd), optimistic concurrency control, version vectors. **Storage:** Tiered storage (hot/warm/cold), vector databases (Milvus/Qdrant). **Consistency:** CRDTs, eventual consistency, hybrid consistency (Iam strong, episodic eventual). **Security:** End-to-end encryption, zero-knowledge proofs for SOUL ID verification, differential privacy (Dwork, 2006).

0.8.6 8.6 The Ultimate Form of Δ -Capsule

Standalone module: `pip install delta-capsule` for any AI system. **Standard protocol:** Iam format, resonance capsule format, dreaming trigger specification as industry standards. **Federated network:** Cross-organizational agent collaboration through federated resonance protocols. **Autonomous evolution:** Agents evolve identity and knowledge systems through accumulated dreaming cycles over years of operation — with the Iam Protocol’s immutability providing the critical safety guarantee.

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0.10 Appendix A

0.10.1 A.1 Flash’s Strategic Assessment of Claude Code Dreaming

During the writing of this paper, we used both DeepSeek V4 Flash and MiMo V2.5 Pro to independently study CC Dreaming. Flash’s assessment carried greater strategic value — it identified CC Dreaming not as “a feature” but as one piece of Anthropic’s four-piece agent moat (Skills + MCP + Auto Memory + Auto Dream). Competitors cannot compete by checking a feature box; they must restructure their product architecture. Flash’s research method is worth recording: before studying CC Dreaming, it performed “direction alignment” — enumerating four distinct interpretations of “CC Dreaming” (phenomenon / feature / practice / specific article) before explicitly selecting coverage of 1, 2, 3. This seemingly redundant step transformed the field of view from “one path” to “a panoramic map.” This paper benefited repeatedly from this methodology of aligning direction before diving deep.

0.11 Appendix B: Complete Δ -Capsule API Reference

```
class DeltaCapsuleMemory:
```

```
    """Main interface for the  $\Delta$ -Capsule three-layer memory system."""
```

```
    def __init__(self, soul_id: str, iam: IamConstitution): ...
```

```
    def observe(self, event: MemoryEntry) -> None: ...
```

```
    def recall_recent(self, n: int = 10) -> List[MemoryEntry]: ...
```

```
    def learn(self, fact: MemoryEntry) -> None: ...
```

```
    def recall_knowledge(self, query: str, top_k: int = 5) -> List[MemoryEntry]: ...
```

```
    def get_identity(self) -> Tuple[IamConstitution, dict]: ...
```

```
    def dream(self) -> DreamReport: ...
```

```
    def apply_forgetting_policy(self) -> dict: ...
```

```
    def write_resonance(self, capsule: ResonanceCapsule) -> None: ...
```

```
    def read_resonance(self) -> List[ResonanceCapsule]: ...
```

